

# Braid Groups

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## Abstract

If you ask for the definition of a braid group, the answer will likely depend on whom you ask. It's not immediately obvious, though, that these different definitions are actually the same. In this talk, we'll explore several ways to define braid groups, convince ourselves that they are indeed equivalent, and get a sense for how we can derive interesting properties of braid groups that have applications to other related groups in low-dimensional topology.

## 1 Braid Groups

### 1.1 Geometric Braids

**Definition 1.1** (Configuration Spaces). Let  $S$  be a surface. We define the *ordered configuration space of  $S$*  as

$$\widehat{C}_n(S) = \left\{ P = (P_1, P_2, \dots, P_n) \in \prod_{i=1}^n \text{int}(S) : P_i = P_j \Rightarrow i = j \right\},$$

endowed with the subspace topology of the product topology on  $\prod_{i=1}^n S$ . Note that the symmetric group  $\Sigma_n$  acts on  $\widehat{C}_n(S)$  by permuting the entries of each  $P$ . We define the *configuration space of  $S$* ,  $C_n(S)$  as the orbit space  $C_n(S)/\Sigma_n$ . It is worth mentioning that the projection  $\widehat{C}_n(S) \rightarrow C_n(S)$  is a regular covering with deck group  $\Sigma_n$ .

This definition simply abstracts the idea of choosing  $n$  distinct points on  $S$ , where the former keeps track of the ordering, while latter cares only about which points we choose. When  $S = D^2$ , we simply write  $\widehat{C}_n$  and  $C_n$  for its configuration spaces.

**Definition 1.2** (Surface Braid Groups). Let  $P \in C_n(S)$ . We define the *geometric braid group* on  $S$  based at  $P$  as

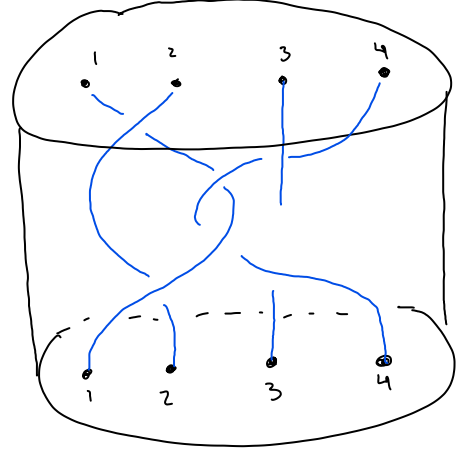
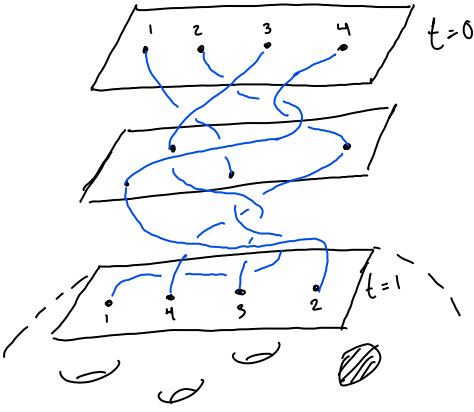
$$B_n(S) := \pi_1(C_n(S), P),$$

and the *geometric pure braid group*

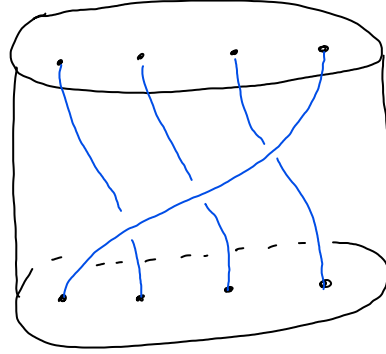
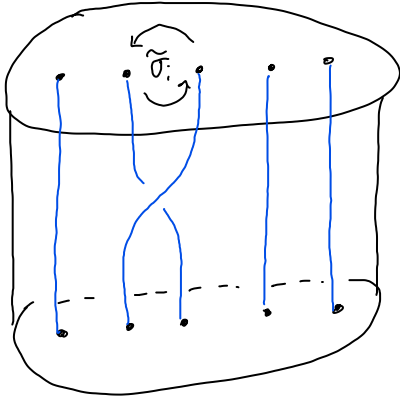
$$PB_n(S) := \pi_1(\widehat{C}_n(S), P).$$

Alternatively, note that we have a map  $\varphi : B_n(S) \rightarrow \Sigma_n$ . We can also define  $PB_n(S) = \text{Ker } \varphi$ .  
Examples of braids:

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Exercise



in terms  
of  $\tilde{\sigma}_i$ 's

Now apply  $\tilde{\sigma}_3 \tilde{\sigma}_2 \tilde{\sigma}_1$

We have a forgetful map  $PB_n(D^2) \rightarrow PB_{n-1}(D^2)$ ; given a geometric pure braid in  $PB_n(D^2)$ , we can forget the first strand to get a pure braid in  $PB_{n-1}(D^2)$ . This map is clearly surjective. Who is the kernel?

More explicitly, we have a fiber bundle  $F : \hat{C}_{n+1} \rightarrow \hat{C}_n$  (again, a forgetful map). To understand the fiber, take for example  $n = 3$ . Fix three points  $p_1, p_2, p_3$  in the interior of the disk (which we may identify with  $\mathbb{R}^2$ ). Under this fiber bundle, what four points can project to these three points? Well, the first three are fixed, so we are free to move the last one, hence this point can lie anywhere in  $\mathbb{R}^2 - \{p_1, p_2, p_3\}$ . In general, we may identify the fiber with  $\mathbb{R}^2 - \{p_1, p_2, \dots, p_n\}$ , which we abbreviate as  $\mathbb{R}_n^2$ . Apply the long exact sequence of homotopy groups to get:

$$\pi_2(\mathbb{R}_n^2) \rightarrow \pi_2(\hat{C}_{n+1}) \rightarrow \pi_2(\hat{C}_n) \rightarrow \pi_1(\mathbb{R}_n^2) \rightarrow \pi_1(\hat{C}_{n+1}) \rightarrow \pi_1(\hat{C}_n) \rightarrow 1,$$

$$1 \rightarrow \pi_2(\hat{C}_{n+1}) \rightarrow \pi_2(\hat{C}_n) \rightarrow F_n \rightarrow PB_{n+1} \rightarrow PB_n \rightarrow 1.$$

Note that  $\hat{C}_1 = \mathbb{R}^2$ , so  $\pi_2(\hat{C}_1) = 1$ , and we get for free that  $\pi_2(\hat{C}_2) = 1$ . By induction,  $\pi_2(\hat{C}_n) = 1$  for all  $n$ . Therefore, we have a short exact sequence

$$1 \rightarrow F_{n-1} \rightarrow PB_n(D^2) \xrightarrow{F_*} PB_{n-1}(D^2) \rightarrow 1.$$

This is a bit of an overkill, but we'll get back to it later.

## 1.2 Mapping Class Groups

Let  $S = S_{g,n}^b$  be a surface, and consider the group  $\text{Homeo}^+(S_{g,n}^b)$  of homeomorphisms that fix  $\partial S$  pointwise, endowed with the compact-open topology. We define the Mapping Class Group

$$\text{Mod}(S) = \pi_0 \text{Homeo}^+(S).$$

This is, isotopy classes of homeomorphisms. Note that homeomorphisms of  $S_{g,n}^b$  preserve the punctures set-wise, but not necessarily pointwise.

We have a homomorphism  $\theta : \text{Mod}(S_{g,n}^b) \rightarrow \Sigma_n$  given by how a homeomorphism permutes the punctures. We define the Pure Mapping Class Group

$$\text{PMod}(S_{g,n}^b) = \text{Ker } \theta.$$

There is a forgetful map

$$\text{Mod}(S_{g,n}^b) \xrightarrow{\mathcal{F}} \text{Mod}(S_g^b)$$

given by ignoring the punctures.

Let  $[\phi] \in \text{Ker } \mathcal{F}$ , i.e. there exists an isotopy  $\phi_t : S_g^b \rightarrow S_g^b$  from  $\phi_0 = \text{Id}$  to  $\phi_1 = \phi$ .

**How does  $\phi_t$  act on the punctures?** It traces out a braid! So  $\text{Ker } \mathcal{F} \subseteq B_n(S_g^b)$ .

Similarly, we have a forgetful map  $\text{PMod}(S_{g,n}^b) \rightarrow \text{Mod}(S_g^b)$ , whose kernel gives pure braids.

In most cases,  $B_n(S) = \text{Ker } \mathcal{F}$ . This is the statement of the Birman Exact Sequence, so we could also define the (pure) braid groups as  $\text{Ker } \mathcal{F}$  (for instance, if we require that  $\chi(S) < 0$ ; this condition is sufficient but not necessary).

## 1.3 Algebraic Braid Groups

If you do geometric group theory, you might care about Coxeter groups, so it might be natural to think about the following group presentation:

$$B_n = \langle \sigma_1, \dots, \sigma_{n-1} \mid \sigma_i \sigma_{i+1} \sigma_i = \sigma_{i+1} \sigma_i \sigma_{i+1}, \sigma_i \sigma_j = \sigma_j \sigma_i, |i - j| > 2 \rangle.$$

We call this the algebraic braid group.

We have yet another homomorphism  $\alpha : B_n \rightarrow \Sigma_n$  given by  $\sigma_i \mapsto (i, i+1)$ , and we define the algebraic pure braid group  $PB_n = \text{Ker } \alpha$ .

A priori, it is not easy to get algebraic properties of  $B_n$ .

## 1.4 How everything connects

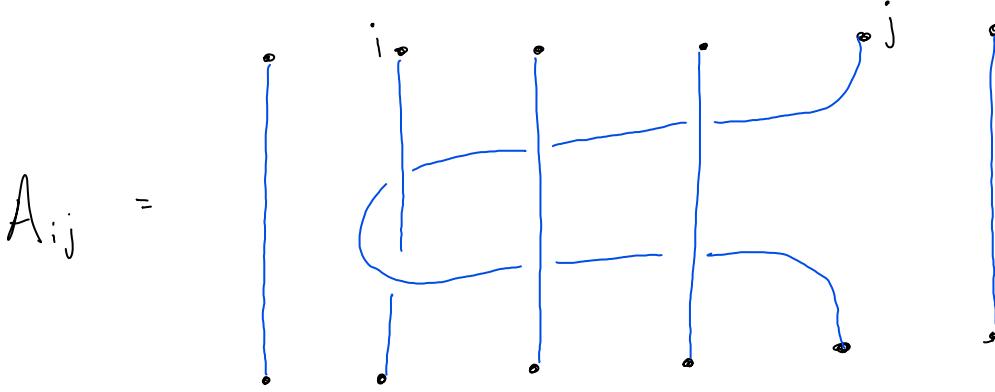
**Theorem 1.3.**  $\text{Mod}(S_{0,n}^1) \cong B_n(D^2) \cong B_n$ .

*Proof.* (Sketch)

1.  $B_n \cong B_n(D^2)$ .

We first find a presentation for  $PB_n$ . What is a good candidate for the set of generators?

Let's cheat using a geometric picture of  $B_n$ .



Let  $A_{i,j} = \sigma_{j-1}\sigma_{j-2}\cdots\sigma_{i+1}\sigma_{i2}\sigma_{i+1}^{-1}\cdots\sigma_{j-2}^{-1}\sigma_{j-1}^{-1}$  ( $1 \leq i < j \leq n$ ). Note that  $[B_n : PB_n] = n!$  (from the short exact sequence). We can give a specific set of  $n!$  coset representatives to find a finite presentation  $PB_n = \langle A_{i,j} \mid R \rangle$  using the Schreier-Reidemeister algorithm. If you are curious, the coset representatives are

$$S = \left\{ \prod_{j=2}^n M_{j,k_j} : j \geq k_k > 1 \right\},$$

where

$$M_{i,j} = \begin{cases} \sigma_{j-1}\sigma_{j-2}\cdots\sigma_i, & j \neq i \\ 1, & j = i \end{cases}$$

We can regard  $PB_{n-1}$  as the subgroup of  $PB_n$  generated by

$$\{A_{i,j} : 1 \leq i < j \leq n-1\}.$$

There's a homomorphism  $\eta : PB_n \rightarrow PB_{n-1}$  given by

$$\eta(A_{i,j}) = \begin{cases} A_{i,j}, & 1 \leq i < j \leq n-1 \\ 1, & j = n \text{ and } 1 \leq i < n. \end{cases}$$

Intuitively,  $\eta$  does what we expect: if the last strand is braided, we forget it; this is some sort of forgetful map.

We can check that  $K_{n-1} = \text{Ker } \eta$  is freely generated by  $A_{1,n}, A_{2,n}, \dots, A_{n-1,n}$ .

Consider the homomorphism  $\rho_n : B_n \rightarrow B_n(D^2)$  given by  $\rho_n(\sigma_i) = \tilde{\sigma}_i$ , where  $\tilde{\sigma}_i$  is the obvious braid. Exercise:  $\rho_n$  is an isomorphism if and only if the restriction of  $\rho_n$  to  $PB_n$ , which we call  $\psi_n$ , is an isomorphism (five lemma).

Now, we have the following commutative diagram

$$\begin{array}{ccccccc} 1 & \longrightarrow & K_{n-1} & \longrightarrow & PB_n & \xrightarrow{\eta} & PB_{n-1} \longrightarrow 1 \\ & & \downarrow \lambda_n & & \downarrow \psi_n & & \downarrow \psi_{n-1} \\ 1 & \longrightarrow & F_{n-1} & \longrightarrow & PB_n(D^2) & \xrightarrow{F_*} & PB_{n-1}(D^2) \longrightarrow 1 \end{array}$$

where  $\lambda_n$  is the restriction of  $\psi_n$  to  $K_{n-1}$ . Now,  $\psi_n$  maps the generators  $\{A_{j,n}, 1 \leq j < n\}$  to a free basis for  $F_{n-1}$ . **Fact:** Free groups are Hopfian. Therefore  $\lambda_n$  is an isomorphism (actually, in this step we can prove that  $K_{n-1}$  is free without having to assume it). Note that  $PB_1 = PB_1(D^2) = 1$ , so  $\psi_1$  is an isomorphism. By induction,  $\psi_n$  is an isomorphism.

2.  $\text{Mod}(S_{0,n}^1) \cong B_n$ .

This result relies on the Birman Exact Sequence.

**Lemma 1.4.** *Let  $P_1, \dots, P_n$  be marked points in  $S_g^b$ . The map  $\mathcal{E} : \text{Homeo}^+(S_g^b) \rightarrow \widehat{C}_n(S_g^b)$  given by*

$$\mathcal{E}(\phi) = (\phi(P_1), \dots, \phi(P_n))$$

*is a fiber bundle with fiber  $\text{PHomeo}^+(S_{g,n}^b)$ .*

We have the long exact sequence

$$\cdots \rightarrow \pi_k(\text{PHomeo}^+(S_{g,n}^b)) \rightarrow \pi_k(\text{Homeo}^+(S_g^b)) \rightarrow \pi_k(\widehat{C}_n(S_g^b)) \rightarrow \pi_{k-1}(\text{PHomeo}^+(S_{g,n}^b)) \rightarrow \cdots$$

The last few terms of the long exact sequence give

$$\cdots \rightarrow \pi_1(\text{Homeo}^+(S_g^b)) \rightarrow PB_n(S_g^b) \rightarrow \text{PMod}(S_{g,n}^b) \rightarrow \text{Mod}(S_g^b) \rightarrow 1.$$

In particular, when  $\chi(S_g^b) < 0$ , we have that  $\pi_1(\text{Homeo}^+(S_g^b)) = 1$ , so we get a short exact sequence

$$1 \rightarrow PB_n(S_g^b) \rightarrow \text{PMod}(S_{g,n}^b) \rightarrow \text{Mod}(S_g^b) \rightarrow 1.$$

We get to keep this fiber bundle when we pass to the quotient  $C_n(S_g^b)$ , so we also have a short exact sequence

$$1 \rightarrow B_n(S_g^b) \rightarrow \text{Mod}(S_{g,n}^b) \rightarrow \text{Mod}(S_g^b) \rightarrow 1.$$

One can prove independently that  $\pi_1(\text{Homeo}^+(D^2)) = 1$  and that  $\text{Mod}(D^2) = 1$ , so the main result has been established:

$$\text{Mod}(S_{0,n}^1) \cong B_n.$$

■

## 2 Properties of braid groups

From now on, when we say braid groups without specifying a surface, we mean  $B_n$ .

- **Braid groups are torsion-free** Recall the topological argument that we used to establish the short exact sequence  $1 \rightarrow F_n \rightarrow PB_n \rightarrow PB_{n-1} \rightarrow 1$ . Writing down more terms of the long exact sequence, we have

$$\pi_k(\mathbb{R}_n^2) \rightarrow \pi_k(\widehat{C}_{n+1}) \rightarrow \pi_k(\widehat{C}_n) \rightarrow \pi_{k-1}(\mathbb{R}_n^2).$$

Clearly  $\pi_k(\mathbb{R}_n^2) = 1$  for  $k \geq 2$ , so  $\pi_k(\widehat{C}_{n+1}) \cong \pi_k(\widehat{C}_n)$ . But  $\pi_k(\widehat{C}_1) = 1$  for  $k > 1$ , so everything becomes trivial. In other words,  $\widehat{C}_n$  is a  $K(\pi, 1)$  space. It follows that  $B_n$  is torsion-free.

- **Braid groups are linear** - Bigelow

## 3 Other braid groups

Consider the braid  $\sigma_1 \sigma_2 \cdots \sigma_{n-2} \sigma_{n-1}^2 \sigma_{n-2} \cdots \sigma_2 \sigma_1$ . Of course, it is not trivial in  $B_n$ , but it becomes trivial in  $B_n(S^2)$ .

Turns out that adding this extra relation gives a presentation for  $B_n(S^2)$ , but it induces torsion. The proof is analogous to the one for  $D^2$  in the case  $n \geq 4$ , but it's true for all  $n$ . The Generalized Birman Exact Sequence applies to  $S^2$ , despite the fact that  $\chi(S^2) = 0$ ; we have  $\text{Homeo}^+(S^2) \simeq \text{SO}(3)$ , so  $\pi_1(\text{Homeo}^+(S^2)) \cong \mathbb{Z}/2\mathbb{Z}$ . However, in the long exact sequence

$$\cdots \rightarrow \pi_1(\text{Homeo}^+(S_{0,n})) \rightarrow \pi_1(\text{Homeo}^+(S^2)) \rightarrow B_n(S^2) \rightarrow \text{Mod}(S_{0,n}) \rightarrow \text{Mod}(S^2) \rightarrow 1,$$

it does happen that  $\pi_1(\text{Homeo}^+(S_{0,n})) = 1$ . Therefore, since  $\text{Mod}(S^2) = 1$ , we still get a short exact sequence:

$$1 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow B_n(S^2) \rightarrow \text{Mod}(S_{0,n}) \rightarrow 1,$$

where  $\mathbb{Z}/2\mathbb{Z}$  corresponds to the center of  $B_n(S^2)$  generated by  $(\sigma_1\sigma_2\cdots\sigma_{n-1})^n$ . Using this short exact sequence, we obtain a presentation for  $\text{Mod}(S_{0,n})$ .

Linearity of  $B_n$  can be pushed to linearity of  $B_n(S^2)$ , which gives linearity of  $\text{Mod}(S_{0,n})$ . A beautiful theorem by Birman and Hilden gives a short exact sequence

$$1 \rightarrow \mathbb{Z}/2\mathbb{Z} \rightarrow \text{Mod}(S_2) \rightarrow \text{Mod}(S_{0,6}) \rightarrow 1,$$

and linearity of  $\text{Mod}(S_{0,6})$  can be pulled back to linearity of  $\text{Mod}(S_2)$ . Whether  $\text{Mod}(S_g)$  is linear for  $g > 2$  remains an open problem.